PROJECT BASED LEARNING (PBL-2)LAB (CSP297)

AI Attendance System

B.TECH 2ndYEAR

SEMESTER: 4th

SESSION: 2024-2025

Submitted By:

Naman Phogat (2024272434)

SECTION: CSE-D

Submitted To

Dr. Ambuj Agarwal

Professor

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SHARDA SCHOOL OF COMPUTING SCIENCE AND ENGINEERING

SHARDA UNIVERSITY, GREATER NOIDA

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Project Title

AI Attendance System

Team / Group Formation:

S. No	Student Name	Roll Number	System ID	Role
1	Naman Phogat	2401010400	2024272434	

Technologies to be used

1. HTML – for creating web page structure
2. CSS – for designing user interface
3. JavaScript – for client-side functionality
4. Python – main programming language
5. Flask – web framework for handling routes and server logic
6. Excel (attendance.xlsx) – used to store student details and attendance records

Hardware Platform

1. A full Development Server
2. Camera
3. Operating System: Windows / macOS / Linux/ Android
4. Editor: Visual Studio Code
5. Browser: Google Chrome

Tools

- **VS Code** – for coding and project development
- **Web Browser (Chrome/Edge)** – for running and testing the system
- **OpenCV** – for face detection using webcam
- **Face Recognition Library** – for identifying student faces
- **OpenPyXL** – for reading and writing Excel file

Problem Statement

*Traditional attendance systems are **manual, time-consuming, and prone to errors and proxy attendance**. There is no secure way to verify student identity, and tracking attendance efficiently is difficult.*

*This project aims to develop a **smart attendance system using face recognition and OTP verification** to ensure **accurate, secure, and automated attendance management**.*

Literature Survey

*Previous studies show that traditional attendance methods are **manual, slow, and allow proxy attendance**. Researchers proposed **Face Recognition based Attendance Systems** to automate attendance using image processing and machine learning. These systems improve **accuracy, security, and efficiency**.*

*Some research also suggests adding **OTP or authentication methods** to increase security and prevent misuse. Based on these studies, the proposed system combines **Face Recognition and OTP verification** to create a **reliable and secure attendance system**.*

Project Description

*The project is a **Smart Attendance System using Face Recognition and OTP Verification**. It has two portals: **Admin Portal** and **Student Portal**.*

*In the **Admin Portal**, the admin logs in and starts the webcam. The system detects and recognizes student faces and automatically marks attendance with **date and time** in the **attendance.xlsx** file.*

*In the **Student Portal**, students log in using **roll number and OTP**. The OTP is sent to the student's **email stored in the Excel file**. After login, students can **view their attendance** in **read-only mode**.*

*The system provides **secure, fast, and accurate attendance management** by combining **face recognition and OTP verification**.*

Project Modules: Design/Algorithm

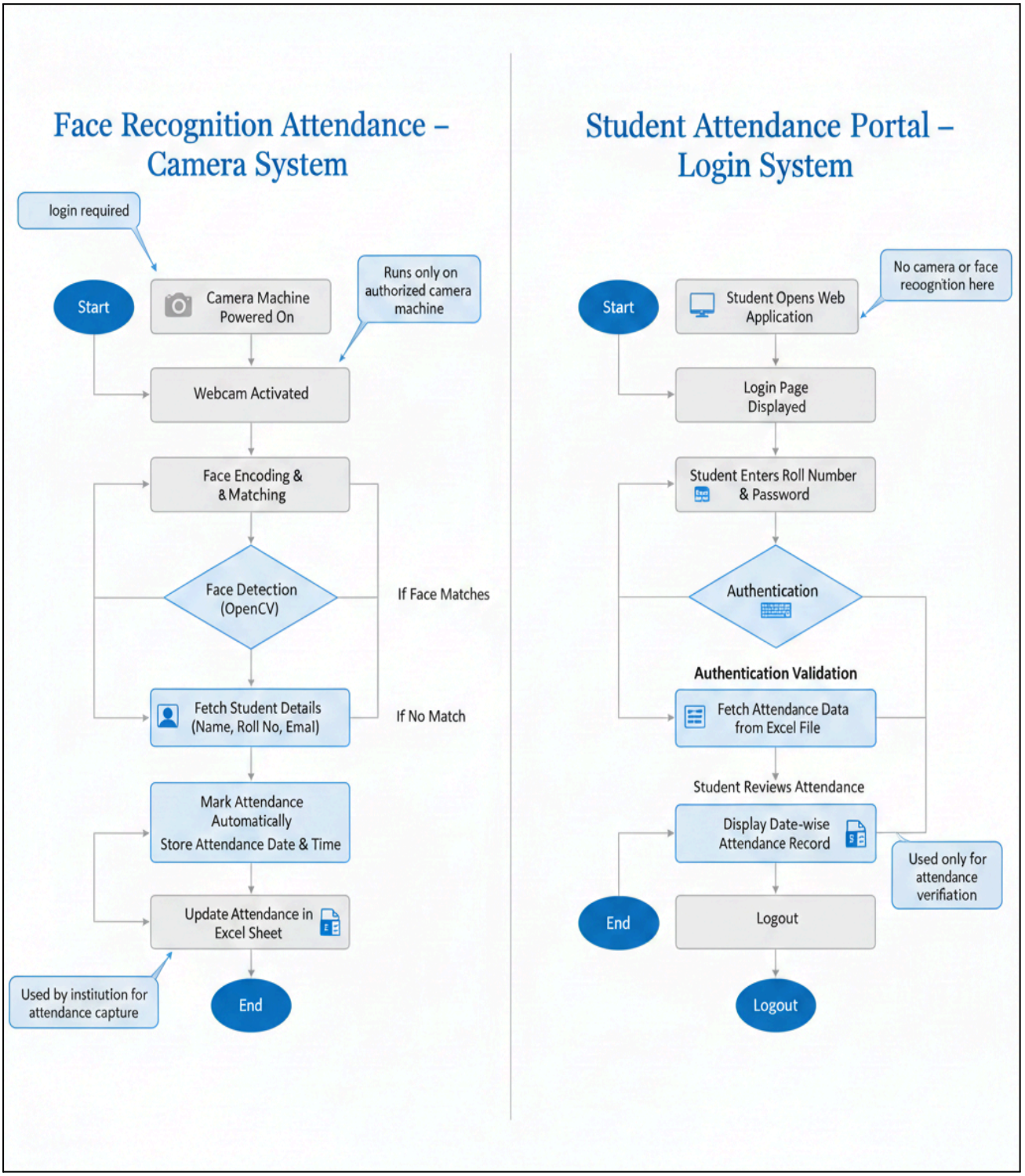
1. Admin Module (Face Recognition)

- *Admin logs into the system.*
- *Webcam starts and captures live images.*
- *The system compares captured face with stored face dataset.*
- *If match is found, attendance is marked in **attendance.xlsx** with **date and time**.*

Algorithm (Face Recognition):

1. *Start webcam*
2. *Capture image frame*
3. *Detect face using OpenCV*
4. *Compare face with stored dataset*
5. *If match found → mark attendance in Excel*
6. *Save date and time*

FlowDiagram



Implementation Methodology

<Guidelines: The process of the whole software system proposed, to be developed, should be mentioned in brief. This may be supported by DFD's / ER Diagram / Class Diagram / Data Models/ Use Case Diagrams/ Flowcharts etc. to explain the flow of the information. Mention how testing of the project will be done and maintenance of the Defect Log>

Future Scope and further enhancement of the Project

- *Integrate **database (MySQL)** instead of Excel for better scalability.*
- *Add **mobile app support**.*
- *Improve accuracy using advanced **deep learning models**.*
- *Add **real-time dashboard** for attendance statistics.*
- *Enable **cloud storage** for remote access.*
- *Add **multi-class attendance management** feature.*

Advantages of this Project

- ***Automatic Attendance** – No manual marking required*
- ***Prevents Proxy** – Face recognition ensures genuine presence*
- ***Secure System** – OTP verification provides safe student login*
- ***Time Saving** – Faster than traditional attendance methods*
- ***Accurate Records** – Attendance stored with date and time*
- ***Easy to Use** – Simple interface for admin and students*
- ***Digital Storage** – Data stored in Excel for easy access*
- ***Low Cost** – Uses basic tools like Python and webcam*
- ***Reliable** – Reduces human errors in attendance management*

Result & Conclusion

The **AI Attendance System** was successfully developed and tested. The system is able to **detect and recognize student faces** using a webcam and automatically mark attendance with **date and time** in the Excel file. The **OTP-based login system** ensures secure access for students to view their attendance in read-only mode.

The project reduces manual effort, saves time, and prevents proxy attendance. The combination of **Face Recognition and OTP verification** makes the system more secure and reliable than traditional attendance methods. Overall, the system provides an **efficient, accurate, and user-friendly solution** for digital attendance management.

Signature

Student Name	Student Sign	Faculty Name	Faculty Sign
Naman Phogat		Dr. Ambuj Agarwal	